Title:

Microneedle-Embedded Drug-Eluting Stent for Targeted MMP-9 Inhibition in Vulnerable Atherosclerotic Plaques

Course:

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Abstract

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Background and Medical Relevance

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Proposed Solution

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Abstract

Atherosclerosis remains a major contributor to cardiovascular disease and mortality worldwide, particularly when plaques become vulnerable and rupture. These vulnerable plaques are characterized by a large lipid core, intense macrophage activity, and a thin fibrous cap prone to degradation. Matrix metalloproteinase-9 (MMP-9), an enzyme secreted by inflammatory macrophages, plays a central role in destabilizing the fibrous cap by degrading structural components such as collagen. Elevated levels of MMP-9 are strongly associated with plaque rupture, thrombosis, and acute cardiovascular events such as myocardial infarction and stroke. Current drug-eluting stents mitigate restenosis through passive drug diffusion but fail to effectively stabilize vulnerable plaques or reach deeply embedded inflammatory cells. Their non-targeted delivery may lead to systemic side effects, insufficient local concentrations, or unintended inflammatory consequences.

To address these limitations, we propose a novel microneedle-embedded drug-eluting stent capable of targeted delivery into the fibrous cap and lipid core of advanced plaques. The stent is fabricated from nitinol, a flexible and biocompatible alloy that exhibits reduced thrombogenicity and strong shape memory, ideal for long-term vascular deployment. Distributed along its surface are arrays of microneedles and drug reservoirs, allowing for controlled, localized drug release regardless of stent rotation or migration. These microneedles penetrate the fibrous cap to deliver MMP-9 inhibitors directly into the site of inflammation, avoiding systemic circulation and minimizing immune clearance.

Two therapeutic agents are employed: morin, a natural flavonoid with anti-inflammatory and antioxidant properties, and JNJ0966, a small-molecule inhibitor that prevents MMP-9 activation through an allosteric mechanism. Morin directly inhibits active MMP-9 and was shown

in both in vitro and in vivo models to suppress endothelial-to-mesenchymal transition, reduce inflammatory signaling, and shrink plaque size. JNJ0966 complements morin by blocking proMMP-9 activation upstream, thus preventing the initiation of the collagen degradation cascade without affecting other MMPs, a significant improvement over prior generations of MMP inhibitors.

Our design builds upon promising data from studies using microneedle drug-eluting balloons, which demonstrated deep tissue penetration (~100 μm), improved drug localization, and superior efficacy in reducing arterial stenosis without increasing vessel injury. The fibrous cap in vulnerable plaques typically ranges from 65–150 μm in thickness, making our microneedle design appropriately dimensioned for safe and effective drug delivery.

The proposed stent also includes potential for theranostic functionality. It can be adapted to monitor local environmental cues such as pH or enzymatic activity and respond by modulating drug release accordingly. This smart-release mechanism enables a more dynamic, responsive approach to therapy without requiring clinician intervention.

Background and Medical Relevance

Atherosclerosis is a chronic disease marked by the buildup of lipids, inflammatory cells, and fibrous tissue within arterial walls, leading to plaque formation that narrows arteries and restricts blood flow. Of particular concern are "vulnerable plaques," which feature a large lipid core, intense macrophage activity, and a thin fibrous cap, making them prone to rupture. Rupture can expose pro-thrombotic material, trigger clot formation, and cause events like heart attacks or strokes. The fibrous cap is gradually weakened by matrix metalloproteinases (MMPs), especially MMP-9, which is secreted by macrophages and degrades structural collagens. Elevated MMP-9 levels are linked to plaque instability and adverse outcomes, making it both a key biomarker and therapeutic target in efforts to prevent rupture and improve treatment of advanced atherosclerotic plaques.[1,2,3]

Conventional drug-eluting stents reduce restenosis by releasing antiproliferative drugs, but fail to address the structural instability of vulnerable plaques. Their reliance on passive diffusion limits drug penetration, making them ineffective for targeting the lipid core or deeply embedded macrophages. They may also release drugs too quickly or diffusely, potentially worsening inflammation. In contrast, microneedle-based systems offer precise, barrier-penetrating drug delivery. Originally developed for transdermal use, microneedles adapted to intravascular devices, like microneedle drug-eluting balloons, have shown enhanced drug deposition in arterial walls without increasing injury or inflammation.[4] These systems demonstrate strong potential for treating complex vascular conditions like atherosclerosis, where deep, localized, and sustained drug release is essential.

Ultimately, the medical relevance of our solution lies in its potential to treat atherosclerosis. Instead of simply mitigating symptoms or performing reactive interventions after a rupture, our design aims to prevent rupture in the first place. This aligns with the broader clinical goal of transitioning cardiovascular care from reactive to preventive, reducing healthcare burden, and improving long-term patient outcomes.

Proposed Solution

The proposed solution is a microneedle-embedded drug-eluting stent that will be used to deliver MMP-9 inhibitors directly into the lipid core of advanced atherosclerotic plaques. The solution begins with using a nitinol stent, as nitinol stents, on average, have 36% less fibrinogen and 63% fewer platelets stuck to them after being used when compared to stainless steel stents[7]. This means fewer clots and a lower risk of inflammation, which is critical from the standpoint of drug elution, where the release of inhibitors in very specific areas of the artery is key. Nitinol is also more flexible and holds its shape better than stainless steel, an important quality when considering that this stent will be releasing a drug over an extended period. Microneedles and each of their respective drug reservoirs will be placed evenly throughout the stent to ensure an equal amount of drug dispersion into the plaque area, and function will be maintained if the stent shifts or rotates within the artery. By utilizing this method of penetrating the fibrous cap using microneedles and targeting macrophage activity beneath, our design aims to stabilize vulnerable plaques and prevent rupture. The stent's diagnostic capabilities consist of detecting changes in the plaque's microenvironment, such as local pH levels and enzymatic activity. When these parameters reach a certain threshold or shift rapidly, the stent is designed to automatically adjust drug release at the target site. This smart-response mechanism enables optimized delivery of MMP-9 inhibitors despite physiological fluctuations, creating a fully autonomous device that functions without the need for clinician intervention.

The microneedles may be fabricated from degradable, biocompatible polymers such as PLA to enable controlled drug release and promote faster endothelial healing. For the implantation procedure, the device is delivered using a small catheter equipped with a balloon at the tip. The catheter is pushed through the vascular system to the site of the atherosclerotic lesion, in which the balloon is then inflated, expanding the stent and keeping it firmly placed against the arterial wall. After the stent is positioned, the balloon is deflated and the catheter system is withdrawn through the same arterial access site.

MMP-9 plays a central role in destabilizing atherosclerotic plaques by degrading extracellular matrix components, particularly type IV and V collagen within the fibrous cap. Elevated MMP-9 levels correlate with larger necrotic core size, plaque instability, and adverse cardiovascular outcomes in both coronary artery disease and acute coronary syndromes.[1] MMP-9 is not only a biomarker but also a therapeutic target. Its inhibition reduces inflammation, neovascularization, the risk of plaque rupture, and can decrease the size of the necrotic core. Currently, however, it is difficult to penetrate into the plaque and effectively access the macrophages within. There is also the risk of potential side effects, such as inhibitors circulating throughout the blood and damaging healthy tissues that require MMP-9 usage for regular remodeling. Thus, by utilizing microneedles, MMP-9 inhibitors can be delivered directly into the fibrous cap, slowing macrophage activity and the degradation of tissue while minimizing immune response. This approach avoids circulation, reducing clearance by blood-filtering organs like the liver and spleen. It also helps bypass immune cells such as macrophages and dendritic cells that might neutralize the drug. To further reduce immune recognition, design strategies like PEGylation, particle size optimization, and limiting repeated exposure can be employed.

To further support our design approach, we will build on the performance of microneedle systems in vascular systems by referencing a study demonstrating the design and efficacy of a microneedle drug-eluting balloon (MNDEB) for drug delivery. Using flexible molding techniques, researchers created 200 µm-high microneedle arrays directly onto the surface of balloon catheters, allowing for strong integrity and minimal risk of detachment during their interfacing with the vascular system. These microneedles were shown to reliably penetrate tissue upon balloon inflation, reaching the tunica media, or the middle of the blood vessel, without compromising the vessel's wall integrity.[3]

In vivo experiments conducted in rabbit iliac arteries revealed that MNDEBs achieved significantly greater drug delivery efficiency compared to conventional drug-eluting balloons.[3] Fluorescent imaging demonstrated a 2.4 times increase in drug penetration into the vessel wall, and a deeper analysis confirmed consistent microneedle insertion depths of approximately 100 µm. This is well within the range needed to bypass the endothelial barrier and access inflammatory macrophages without breaching deeper layers.[3] This enhanced penetration allowed for more localized and sustained drug exposure.

Therapeutic efficacy trials in a rabbit atherosclerosis model indicated that MNDEB treatment resulted in a significant reduction in both diameter and area stenosis when compared to traditional DEBs. These findings validate the concept that mechanical insertion of microneedles changes the precision and effectiveness of drug delivery within complex vascular geometries. Given that the fibrous cap of advanced atherosclerotic plaques typically ranges between 65–150 µm, the microneedle height and penetration profile reported in this study align closely with our stent design, reinforcing its feasibility for safe and effective plaque-targeted therapy.

Histological analysis, or analysis of tissue structures, revealed no significant differences in vessel injury scores, inflammation markers, or endothelialization between MNDEB-treated vessels and those treated with conventional balloons.[3] This safety profile highlights the biocompatibility and mechanical appropriateness of microneedle-based delivery even in diseased vascular environments. These findings strongly support the integration of microneedle technology into our stent design to achieve both high efficacy and mechanical safety within atherosclerotic vessels.

One of the main compounds we looked at to inhibit MMP-9 was morin, a natural compound found in white mulberry known to contain anti-inflammatory and antioxidant properties. Human umbilical vein endothelial cells are a common type of cell that line blood vessels, making them a viable subject for mimicking EndMT and testing the effects of morin. Researchers exposed the cells for over 48 hours to 10 ng/mL of transforming growth factor-β1 (TGF-β1), a signaling molecule known to cause endothelial-to-mesenchymal transition (EndMT). After being exposed to TGF-β1, MMP-9 and Notch-1 signaling levels increased, with both pathways promoting EndMT and inflammation. The result of treating the human umbilical vein endothelial cells with morin showed a dose-dependent response: 50 μmol/L of morin suppressed both MMP-9 and Notch-1 signaling in TGF-β1-induced EndMT, preventing EndMT. The study then increased MMP-9 levels independently, which triggered Notch-1 signaling and reversed morin's effect, proving morin works by inhibiting MMP-9.

Researchers utilized mice that are genetically modified to lack the gene apolipoprotein E, which helps break down fats. These mice easily developed atherosclerosis given their high-fat diet, making them ideal models for observing arterial plaque formation. A sample of mice was given oral doses of 50 mg/kg of morin over four weeks, resulting in notable reductions of aortic intimal hyperplasia and plaque formation by suppressing EndMT through MMP-9 inhibition. The final portion of the study focused on molecular docking and microscale thermophoresis to understand how morin physically interacts with MMP-9. Morin was found to have a strong binding affinity for MMP-9, demonstrating the high likelihood that morin inhibits or weakens MMP-9 activity directly[5]. The success of the in vivo study highlights morin's therapeutic potential by demonstrating reductions in both intimal hyperplasia and arterial plaque formation. With proper dosing and delivery, morin could serve as a biocompatible, phytotherapeutic alternative or complement to synthetic MMP-9 inhibitors.

In addition to morin, our therapeutic strategy incorporates JNJ0966, a highly selective small-molecule inhibitor that represents a breakthrough in MMP-9 targeting. Unlike previous generations of MMP inhibitors that broadly targeted the catalytic zinc-binding domain-leading to dose-limiting musculoskeletal toxicity in clinical trials[4]. JNJ0966 operates through a novel allosteric mechanism. It specifically binds to a regulatory pocket adjacent to Arg-106 in the proMMP-9 zymogen, preventing the proteolytic cleavage and structural reorganization required for activation[4]. This unique binding mode confers exceptional specificity, showing no inhibitory activity against MMP-1, -2, -3, or -14 catalytic domains while effectively blocking proMMP-9 activation with an IC_{50} of approximately 440 nM[4].

Within atherosclerotic plaques, JNJ0966's upstream inhibition strategy offers distinct advantages for stabilizing vulnerable lesions. By preventing proMMP-9 activation rather than inhibiting an already-active enzyme, it halts the proteolytic cascade before collagen degradation can compromise the fibrous cap's structural integrity. Our microneedle-embedded stent design ensures precise delivery of JNJ0966 directly into the lipid-rich necrotic core, where the drug can diffuse radially to reach MMP-9-producing macrophages. The stent's controlled release mechanism maintains therapeutic concentrations over time, addressing the chronic nature of plaque progression while avoiding the pitfalls of systemic MMP inhibition. When combined with morin's direct inhibition of active MMP-9 and antioxidant effects, this dual-pathway approach provides comprehensive protection against plaque destabilization. The specificity of JNJ0966 for the MMP-9 zymogen activation process, validated through extensive biochemical and structural studies, makes it an ideal candidate for our targeted therapeutic strategy aimed at preventing acute coronary events.

Summary & Conclusion

Our solution outlines a novel microneedle-embedded drug-eluting stent that is designed to deliver matrix metalloproteinase-9 (MMP-9) inhibitors such as morin and JNJ0966 directly into the lipid core of atherosclerotic plaques. In these atherosclerosis plaques, an excess of MMP-9 destabilizes the plaque, thinning out the fibrous cap and increasing the risk of rupture. Made of nitinol, our novel stent takes advantage of its ductile and biocompatible properties, incorporates evenly distributed micro-needles to allow for localized drug release into the fibrous cap of an atherosclerotic plaque region. We referenced research about microneedle drug-eluting balloons that supported conclusions on how enhancing drug delivery efficiency without increasing the risk of vascular injuries is possible with this kind of stent. By delivering morin, a natural compound, into the fibrous cap of this plaque as our MMP-9 inhibitor, we found that its anti-inflammatory characteristics reduce MMP-9 activity, attempting to stabilize the region that has been affected by atherosclerotic plaques. Seeing as our novel solution innovates an already established vascular procedure without increasing risk factors towards the patient, we believe that our proposed solution offers a promising and targeted approach towards treating atherosclerotic plaques. Through the integration of microneedles for precise drug delivery and the use of morin for effective MMP-9 inhibition, our proposed solution looks to not only stabilize the atherosclerotic plaques but also reduce the risk of ruptures to improve post-op patient quality of life.

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Illustrations

Fig. 1: Schematic representation of how our novel microneedle-embedded drug-eluting stent would deliver MMP-9 inhibitors directly into the lipid core of atherosclerotic plaques

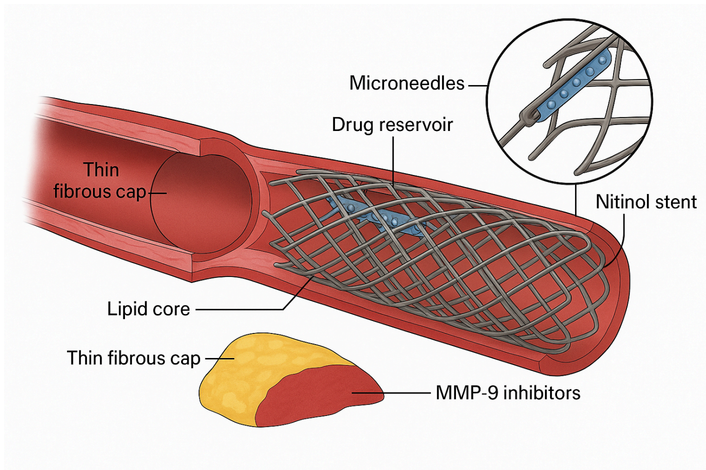

Fig. 2: Basic Stent Design

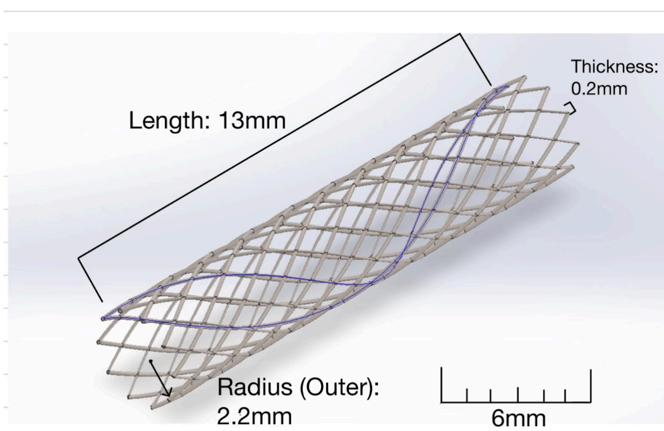

Fig. 3: Stent with Microneedle (Cone) and Drug Reservoir (Rectangular Prism)